

withdrawal rate of 5.9 percent of initial assets. Why was the withdrawal rate barely over 4 percent?

We saw earlier that a 1.3 percent return assumption is needed to make the 4 percent rule work. The further amplifying effects of sequence risk on investment volatility made it seem like the compounded return was only 1.3 percent for retirees that year, instead of the actual 4.2 percent. The difference between the 4.15 percent and 5.9 percent spending rates is another illustration of the specific impact of market returns sequence in that thirty-year period. The early part of the 1966 retiree's retirement was a tough time, with market losses in 1966, 1968, and 1973–74. This early period set the course for sustainable spending and caused spending to fall below what was implied by the average over the whole period. The markets boomed in the second half of this retirement period, but by then it was too late. The portfolio was already on an unsustainable trajectory, leading to the lowest withdrawal rate in this historical period.

Bengen showed that, historically, a 4 percent initial withdrawal turned out to be much more realistic than higher numbers found when ignoring market volatility. Hence, 4 percent became the rule of thumb for retirement withdrawals.

For the following discussion, I mostly will use the same assumptions as Bengen's original research, with one exception: I assume retirees make their withdrawals at the start of each year, while Bengen assumes end-of-year withdrawals. I think withdrawals at the start of the year are more realistic, since retirees need the funds in order to be able to spend them, and this assumption causes my SAFEMAX to be 4.03 percent, compared to Bengen's 4.15 percent.